University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of CSE

Term Final Examination, Spring-2024

Course Title: Coordinate Geometry and Vector Analysis

Course Code: MAT261

Marks: 50 **Time: 3 Hours**

(Answer all the questions)

1. (a) Test the nature of the conics given by the following equations: [04]

   i) $x^2 - 6xy - 2y^2 - 4y + 6 = 0$

   ii) $4x^2 + 9y^2 + 36y - 8x - 31 = 0$

   (b) Reduce the equation $5x^2 - 24xy - 5y^2 + 58y + 4x - 59 = 0$ to its standard form. [06]

2. (a) Define direction cosines and direction ratios of a line. Find the direction cosines l, m, n of two lines which are connected by the relations, [05]

   $$l - 5m + 3n = 0,\ \ 7l^2 + 5m^2 - 3n^2 = 0.$$

   (b) What is plane? Find the equation of the plane passing through the points (1, 0, -1), (2, 1, 3) and is perpendicular to the plane $2x + y + z - 1 = 0$. [05]

3. (a) The following forces act on a particle P. Find the magnitude of the resultant vectors [05]

   $\bar{F}_1 = 2\hat{\imath} + 3\hat{\jmath} - 5\hat{k}, \bar{F}_2 = -5\hat{\imath} + \hat{\jmath} + 3\hat{k}, \bar{F}_3 = \hat{\imath} - 2\hat{\jmath} + 4\hat{k}$ & $\bar{F}_4 = 4\hat{\imath} - 3\hat{\jmath} - 2\hat{k}$.

   (b) What is dot and cross product? Find the directional derivatives of $\phi = 4xz^3 - 3x^2y^2z$ at (2,-1, 2) in the direction $2\underline{i} - 3\underline{j} + 6\underline{k}$. [05]

4. (a) Define gradient and Curl. Find $\nabla\varphi$ if $\varphi = ln|\mathbf{r}|$ and also prove that $\vec{w} = \frac{1}{2} curl\ \vec{v}$. [06]

   (b) A particle moves along the curve $x = 2t^2, y = t^2 - 4t, z = 3t - 5$, where t is the time. Find the components of its velocity and acceleration at time t=1 in the direction $\underline{i} - 3\underline{j} + 2\underline{k}$. [04]

5. (a) If $\varphi = 2xyz^2, \underline{F} = xy\underline{i} - z\underline{j} + x^2\underline{k}$ and C is the curve $x = t^2, y = 2t, z = t^3$ from $t = 0$ to $t = 1$, evaluate the line integrals [06]

   (i) $\int \varphi d\underline{r}$ (ii) $\int \underline{F} \times d\underline{r}$

   (b) State Green's Theorem. Verify the Green's theorem in the plane for $\oint_C (xy + y^2)\, dx + x^2\, dy$ where C is the closed curve of the region bounded by y = x and y = x^2 . [04]